

EXAMINATION OF THE EFFECT OF ELEVATED NaCl CONCENTRATIONS ON THE BURDEN OF THE ECTOPARASITES TRICHODINIDS AND DACTYLOGYRIDS ON PIKEPERCH (*SANDER LUCIOPERCA*) MAINTAINED IN RECIRCULATING AQUACULTURE SYSTEMS (RAS)

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Introduction

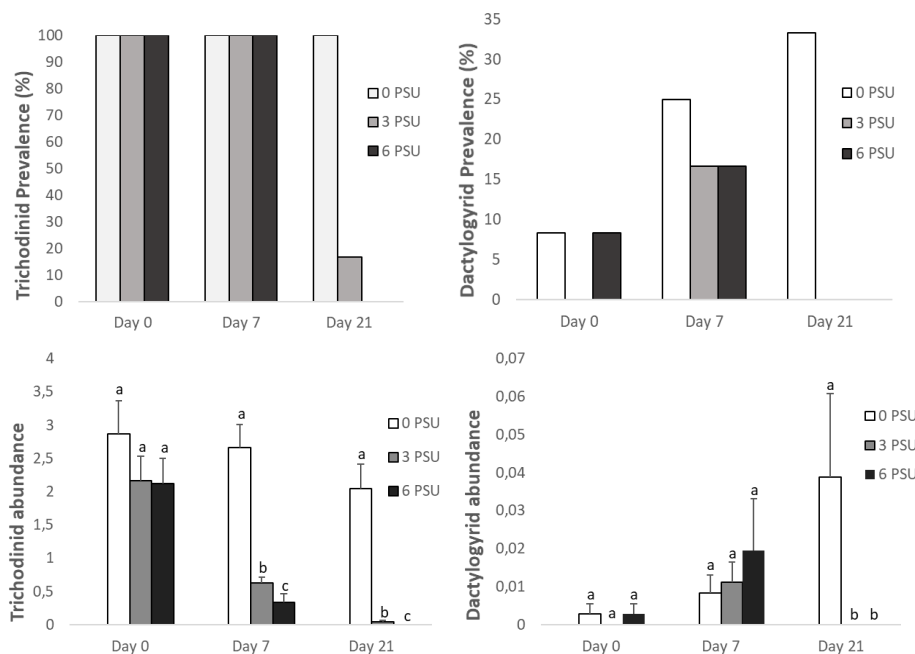
Pikeperch is a high valued species and one of the favorite food fishes in Europe. Declining wild catches and increasing demand have raised the popularity to intensify production techniques of this species using recirculating aquaculture systems (RAS). However, RAS possesses disadvantages, as high stocking densities increase the risk of disease transmission of pathogens and ectoparasites (Bregnballe, 2015; Németh et al., 2013). One of the most innocuous treatments for freshwater fish in a closed aquaculture system is using low concentrations of sodium chloride (NaCl) as prolonged immersion (Noga, 2010). This study was conducted to evaluate the effect of elevated NaCl concentrations on the burden of common ectoparasites in pikeperch, protozoan Trichodinids and monogenean Dactylogyrids, maintained in RAS.

Materials and Methods

Pikeperch (120.5 ± 15.7 g) were brought from overwintering conditions in Lake Sacrow, Potsdam-Sacrow, Germany. The treatments consisted of a negative control stocked with uninfected pikeperch, a positive control stocked with infected pikeperch, and two treatment groups stocked with infected pikeperch and addition of NaCl 3 PSU and 6 PSU, respectively. Both controls were kept without NaCl application (0 PSU). All treatments were performed in three replicates in four units of RAS for 21 days. Each RAS consisted of three rearing tanks (320 L) and a filtration unit (120 L). Each RAS was stocked with 57 fish (19 fish/tank). On days 0, 7, and 21 of the experiment, four fish per tank were sacrificed (total 12 fish/treatment) for ectoparasite analysis. The mucus from several areas of skin around the dorsal fin, anal fin, ventral fin, pectoral fin was swabbed, and the second raker of the right gill was dissected for examination. The smear of mucus and gill samples were examined within six windows of observation at 100 fold magnification under a light microscope (window per view = 0.031 mm^2).

Results

Regarding performance parameters, there were no significant differences in specific growth rate (SGR), feed conversion ratio (FCR), survival rate (SR), final biomass, and condition factor (CF) among all treatments ($p \leq 0.05$). At the beginning and day 7 of the experiment, ectoparasites Trichodinids and Dactylogyrids were found in the positive control (0 PSU), 3 PSU, and 6 PSU treatments. In contrast, there were no ectoparasites found in the negative control. At the end of the experiment, we observed a declining trend of prevalence and abundance of Trichodinids (by 83.3% and 98.15% in 3 PSU and both 100% in 6 PSU treatments) and Dactylogyrids (by 100% both in 3 PSU and 6 PSU). Oppositely, there was a stable trend of Trichodinids prevalence (100%) and an increasing trend of Dactylogyrids prevalence (up to 25%) in the positive control (0 PSU) from the beginning to the end of the experiment.



Discussion

Among all treatments, the application of 6 PSU NaCl was the most effective in controlling the ectoparasites, Trichodinids and Dactylogyrids. A relatively similar result based on Németh et al. (2013), the continuous application of 5 PSU NaCl was able to eliminate *Trichodina* sp. within two weeks of application. Based on this finding it might be possible that 6 PSU is sufficient to eliminate Trichodinids. Concentrations below these ranges (e.g. 3 PSU) are probably capable of reducing, not completely removing all Trichodinids on the host. A possible mechanism how NaCl can remove ectoparasites is initiating an osmotic pressure difference that is harmful (Kirschner, 2004) or removing an excess of mucus and of debris being associated with ectoparasite infestations (Noga, 2010).

Conclusion

The application of elevated NaCl concentrations (3 PSU and 6 PSU) can reduce the prevalence and abundance of the ectoparasites, Trichodinids and Dactylogyrids. Moreover, there were no ectoparasites found on pikeperch in the 6 PSU treatment after three weeks of the NaCl prolonged application. Though further researches are needed, our findings provide an effective treatment method for controlling both ectoparasites, Trichodinids and Dactylogyrids, on pikeperch maintained in RAS.

References

- Bregnballe, J. (2015). A Guide to Recirculation Aquaculture: An introduction to the new environmentally friendly and highly productive closed fish farming systems. FAO and Eurofish Report, 100.
- Kirschner, L. B. (2004). The mechanism of sodium chloride uptake in hyperregulating aquatic animals. *Journal of Experimental Biology*, 207(9), 1439–1452.
- Németh, S., Horváth, Z., Felföldi, Z., & Beliczky, G. (2013). The use of permitted ectoparasite disinfection methods on young pike-perch (*Sander lucioperca*) after transition from over-wintering lake to RAS. *AACL Bioflux* 6(1), 1–11.
- Noga, E. J. (2010). *Fish Disease: Diagnosis and Treatment*, Second Edition (2nd ed.). North Carolina, US: John Wiley & Sons, Inc.